

Abdul Wahid

ICSFULESTATION- Let's-Defend CTF Investigation

DIGITAL FORENSICS & CYBER-THREAT INTELLIGENCE

Contents

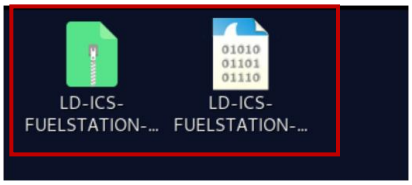
Challenge Overview 4
Question 1 — Initial Connection Timestamp (SYN Only) 5
Approach 5
Steps Performed 5
Question 2 — First Full SYN Packet Timestamp 6
Approach 6
Steps Performed 6
Question 3 — Attacker's Source IP Address 7
Approach 7
Steps Performed 8
Question 4 — Destination Port (Service Port) 8
Approach 9
Steps Performed 9
Question 5 — Custom/High Port Used by the Protocol 10
Approach 10
Steps Performed 10
Question 6 — ICS Protocol / Vendor Identification 11
Approach 11
Steps Performed 11
Question 7 — Facility / Site Name 13
Approach 13
Steps Performed 13
Question 8 — Tank / Sensor Identifier 14
Approach 15
Steps Performed 15
Question 9 — Scheduled Delivery / Operation Time 16
Approach 16
Steps Performed 16
Question 10 — Current Fill / Operational Status 18
Approach 18
Steps Performed 18
Question 11 — Delivery Start Timestamp 19
Approach 19
Steps Performed 19
Question 12 — Delivery End Timestamp 20

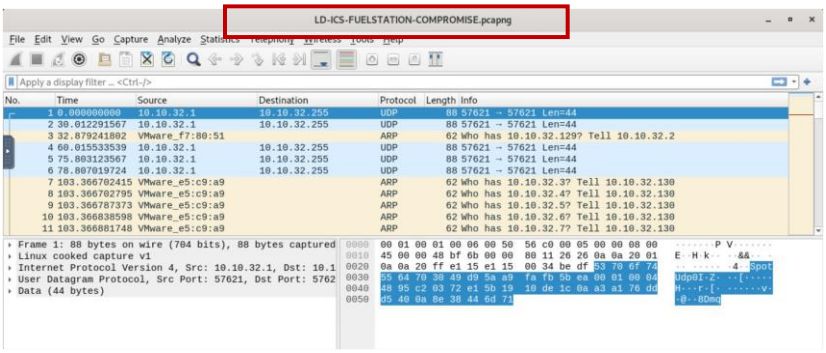
Approach 20
Steps Performed 20
Question 13 — Fuel Level / Volume Change 21
Approach 21
Steps Performed 21
Question 14 — Fuel Type / Product Classification 22
Approach 22
Steps Performed 22
Question 15 — DHCP Hostname / Site Identifier 23
Approach 23
Steps Performed 23
Question 16 — Ransom Demand / Extortion Amount 24
Approach 24
Steps Performed 24
Question 17 — Attacker Contact / C2 Email Address 26
Approach 26
Steps Performed 26

Challenge Overview

This report documents the forensic investigation of the ICS FuelStation challenge on LetsDefend. The challenge simulates a real-world scenario where an attacker compromised a critical industrial endpoint — an Automatic Tank Gauge (ATG) system at a fuel distribution site — and conducted C2 communications over a non-standard protocol. The analyst is tasked with examining captured network traffic to uncover attacker activity, identify the threat, and extract key indicators of compromise.

> *Challenge File: /root/Desktop/LD-ICS-FUELSTATION-COMPROMISE.zip → LD-ICS-FUELSTATION-COMPROMISE.pcap — Opened in Wireshark for all analysis tasks below.*





Question 1 — Initial Connection Timestamp (SYN Only)

When did the first SYN packet (connection initiation) occur in the capture?

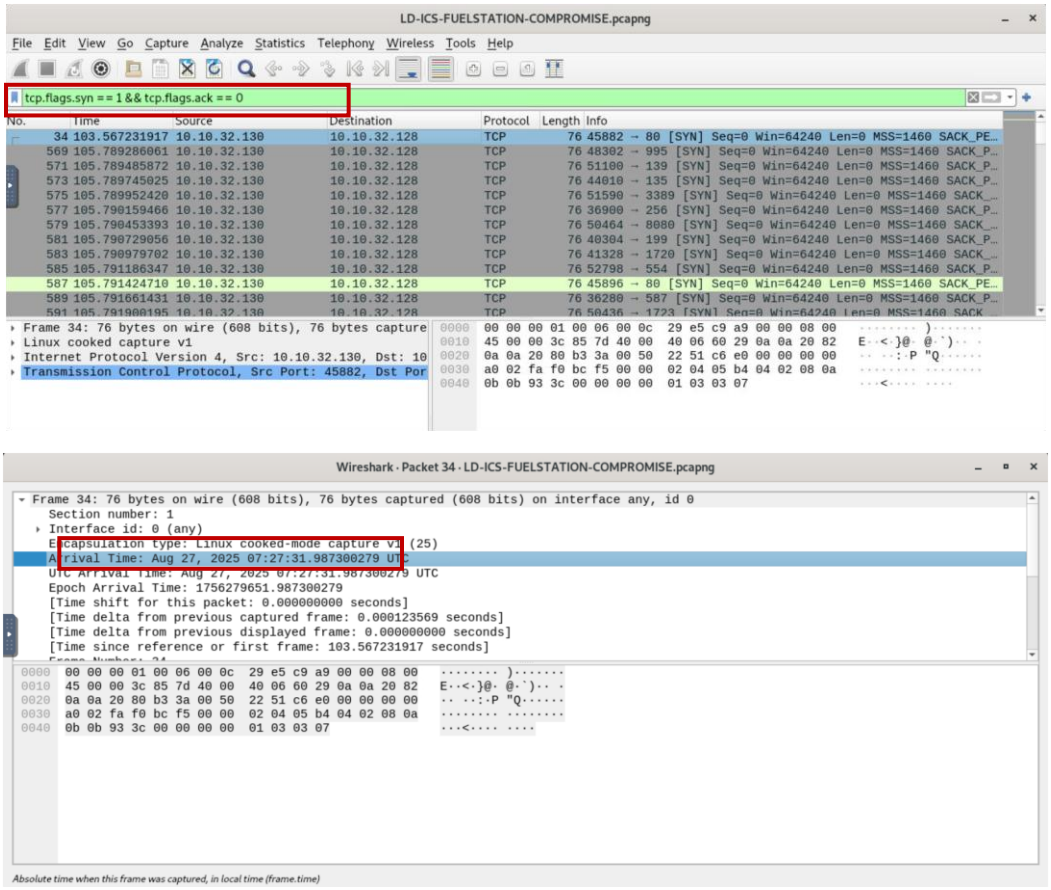
> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

Approach

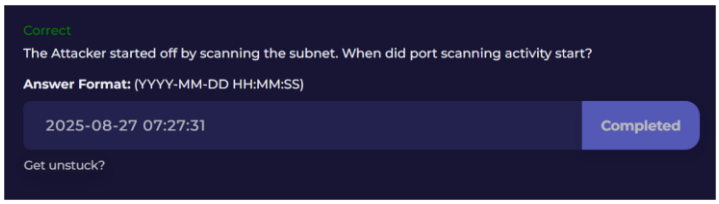
In network forensics, a SYN-only packet (tcp.flags.syn==1 && tcp.flags.ack==0) represents the very first step of the TCP three-way handshake — the client initiating a connection. Identifying the earliest such packet establishes the attacker's first contact time with the target system.

Steps Performed

- Connected to the challenge machine and opened the file LD-ICS-FUELSTATION-COMPROMISE.pcap in Wireshark.
- Applied the Wireshark display filter: tcp.flags.syn == 1 && tcp.flags.ack == 0
- Sorted results by time and read the Arrival Time from the first matching frame.



Wireshark filtered Output showing first SYN packet & its arrival time



✓ **Answer: 2025-08-27 07:27:31**

Question 2 — First Full SYN Packet Timestamp

What is the arrival time of the first SYN packet (tcp.flags.syn == 1)?

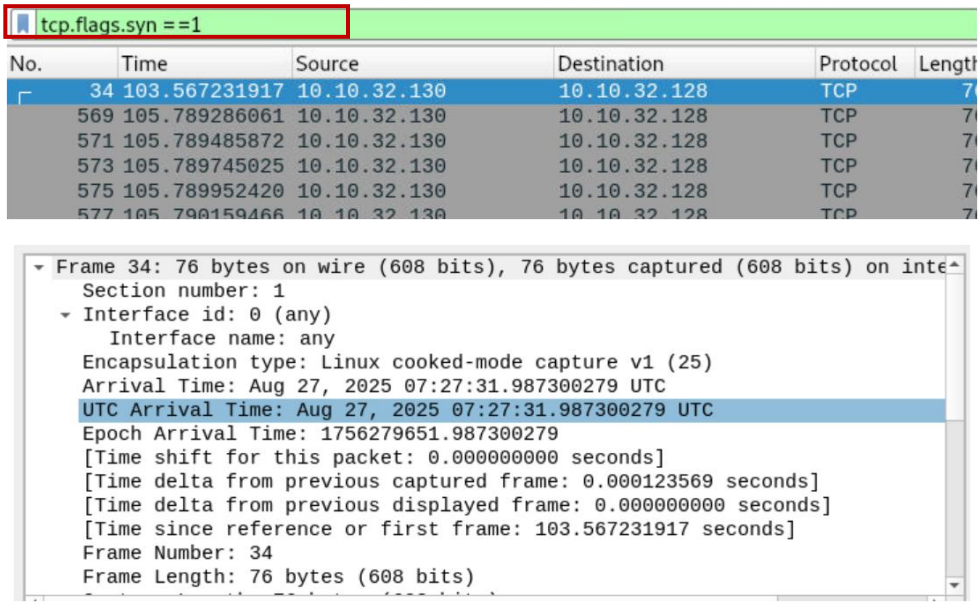
> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

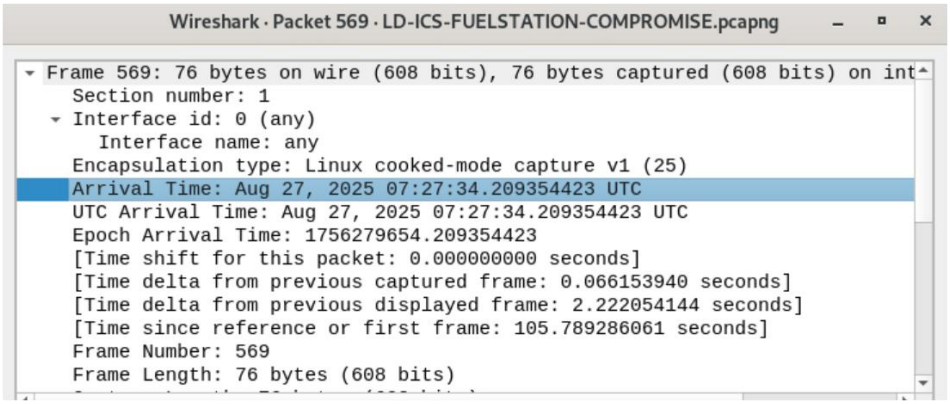
Approach

Filtering for all SYN packets (including SYN-ACK) provides a broader picture of initial connection events. The first packet in this filter shows the timestamp when the server responded or the full handshake was captured, useful for timeline correlation.

Steps Performed

- Applied the Wireshark filter: tcp.flags.syn == 1
- Navigated to the Frame section in the packet details pane.
- Recorded the Arrival Time value from the first matching frame.





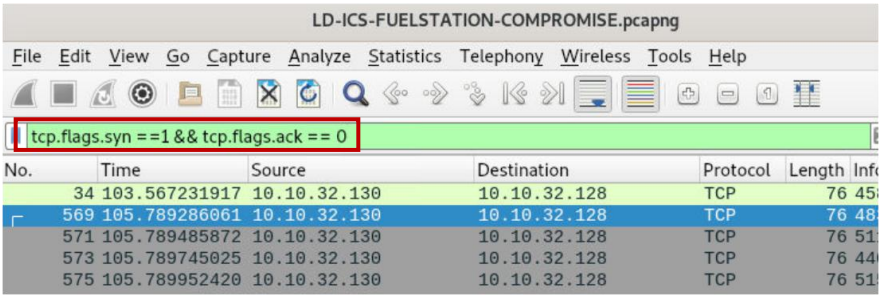


✓ **Answer: 2025-08-27 07:27:34**

Question 3 — Attacker's Source IP Address

What is the IP address of the system that initiated the connection (SYN sender)?

> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

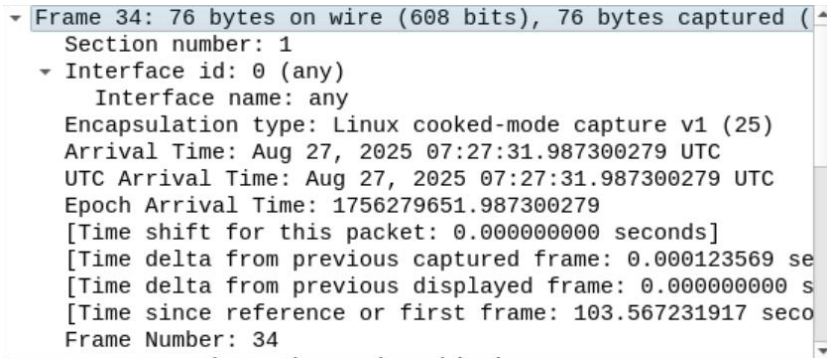


Approach

The source IP of a SYN-only packet identifies the initiating party of the TCP connection. In incident response, pinpointing this address is critical for attribution, threat actor mapping, and blocking malicious hosts at the firewall level.

Steps Performed

- Applied the filter: tcp.flags.syn == 1 && tcp.flags.ack == 0
- Inspected the Source IP column of the first filtered packet.
- Noted the IPv4 address as the connection originator.



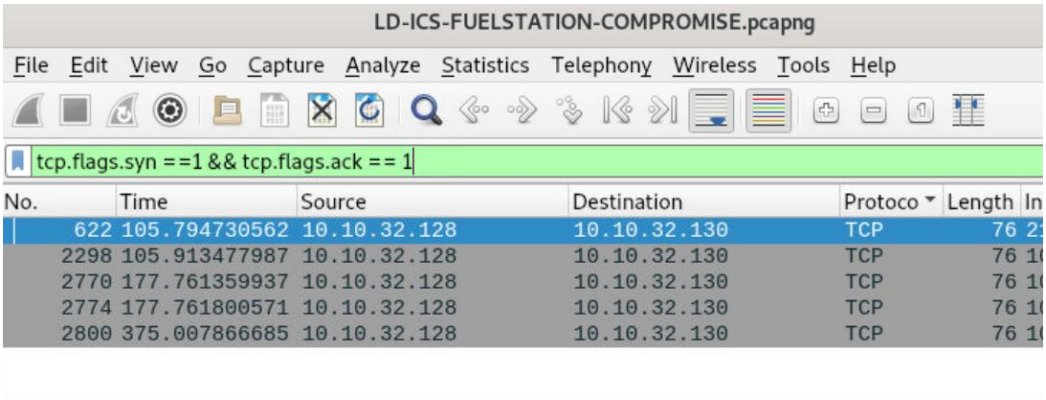


✓ **Answer: 10.10.32.130**

Question 4 — Destination Port (Service Port)

What destination port was targeted in the SYN-ACK response?

> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

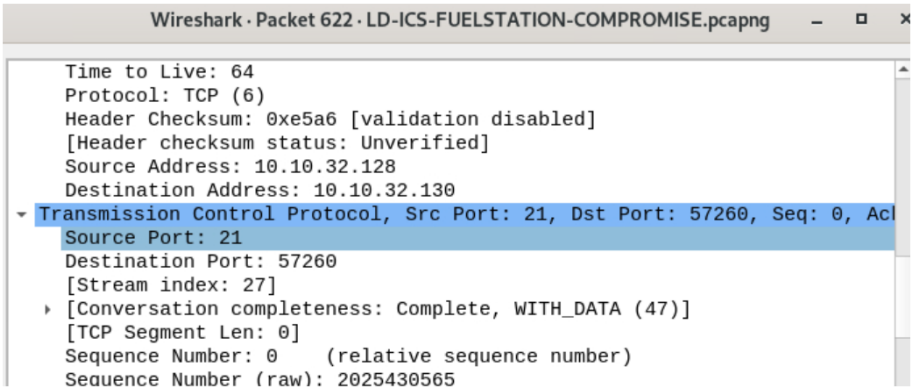


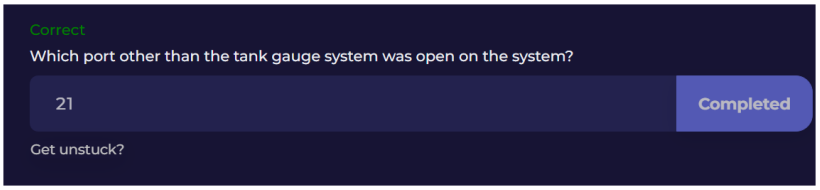
Approach

SYN-ACK packets (tcp.flags.syn==1 && tcp.flags.ack==1) are sent by the server to acknowledge an incoming connection. The destination port in SYN-ACK traffic reveals which service was actively listening and accepted the connection — in this case, an ICS-related service.

Steps Performed

- Applied the filter: tcp.flags.syn == 1 && tcp.flags.ack == 1
- Examined the Destination Port column in the filtered packet results.
- Identified port 21 as the responding service port.



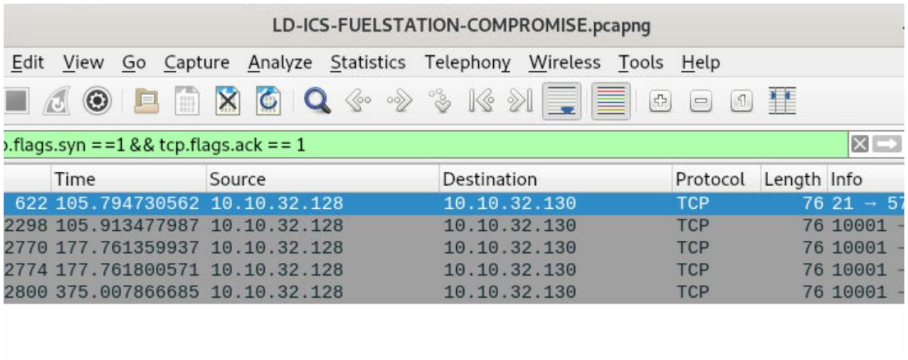


✓ **Answer: 21**

Question 5 — Custom/High Port Used by the Protocol

What non-standard (high) port was used for ICS communications?

> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

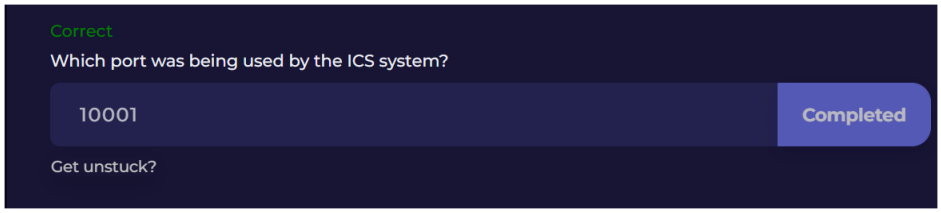


Approach

Industrial Control Systems (ICS) often expose services on non-standard, high-numbered ports to avoid detection. Discovering this port is crucial for understanding the attacker's communication channel and helps analysts pivot into protocol-specific analysis.

Steps Performed

- Continued reviewing SYN-ACK filtered traffic: tcp.flags.syn == 1 && tcp.flags.ack == 1
- Identified a second responding port beyond standard services.
- Noted port 10001 as the custom ICS communication port.

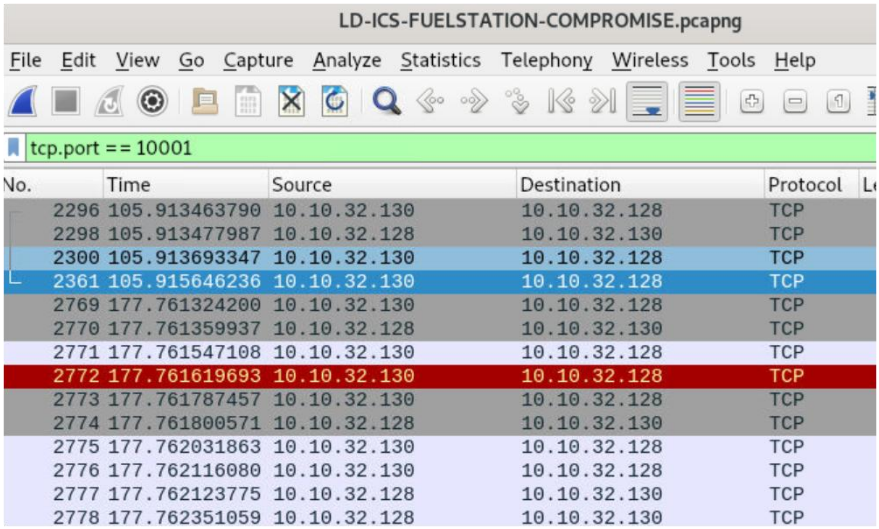


✓ **Answer: 10001**

Question 6 — ICS Protocol / Vendor Identification

What ICS vendor or protocol was communicating over the discovered port?

> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

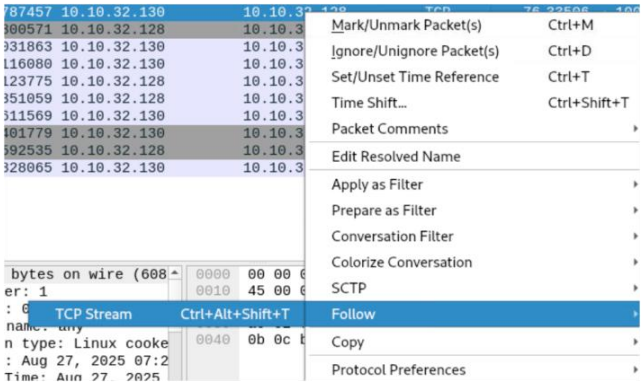


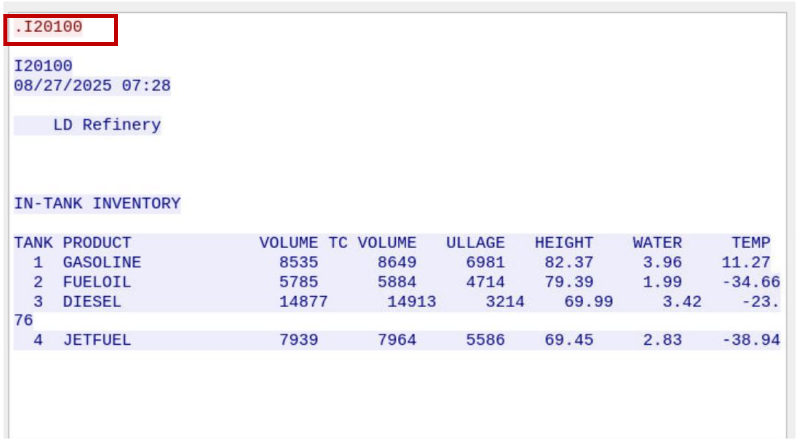
Approach

By filtering traffic on the identified ICS port (10001) and analysing the payload or protocol dissector, analysts can identify the vendor-specific protocol in use. Wireshark's automatic protocol dissection often reveals the protocol name directly in the packet list.

Steps Performed

- Applied the filter: tcp.port == 10001
- Examined the protocol column and packet bytes in matching packets.
- Identified the Veeder protocol (Veeder-Root ATG — Automatic Tank Gauge) in the payload.





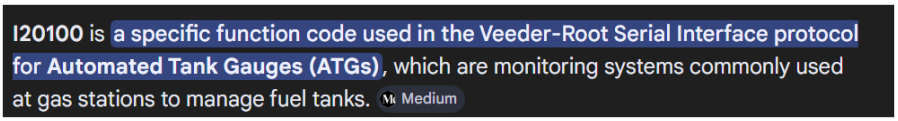
```
.I20100

I20100
08/27/2025 07:28

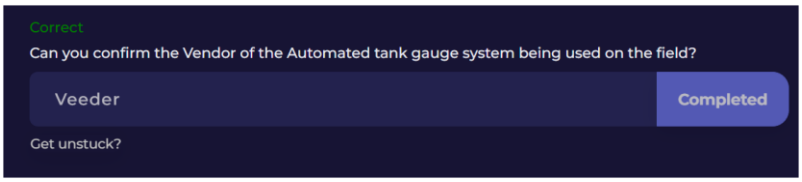
    LD Refinery


IN-TANK INVENTORY

TANK PRODUCT               VOLUME TC VOLUME   ULLAGE   HEIGHT    WATER     TEMP
  1  GASOLINE                 8535      8649     6981    82.37     3.96    11.27
  2  FUELOIL                  5785      5884     4714    79.39     1.99    -34.66
  3  DIESEL                  14877     14913     3214    69.99     3.42    -23.
76
  4  JETFUEL                  7939      7964     5586    69.45     2.83    -38.94
```



I20100 is a specific function code used in the Veeder-Root Serial Interface protocol for Automated Tank Gauges (ATGs), which are monitoring systems commonly used at gas stations to manage fuel tanks. Medium



Correct

Can you confirm the Vendor of the Automated tank gauge system being used on the field?

Veeder | Completed

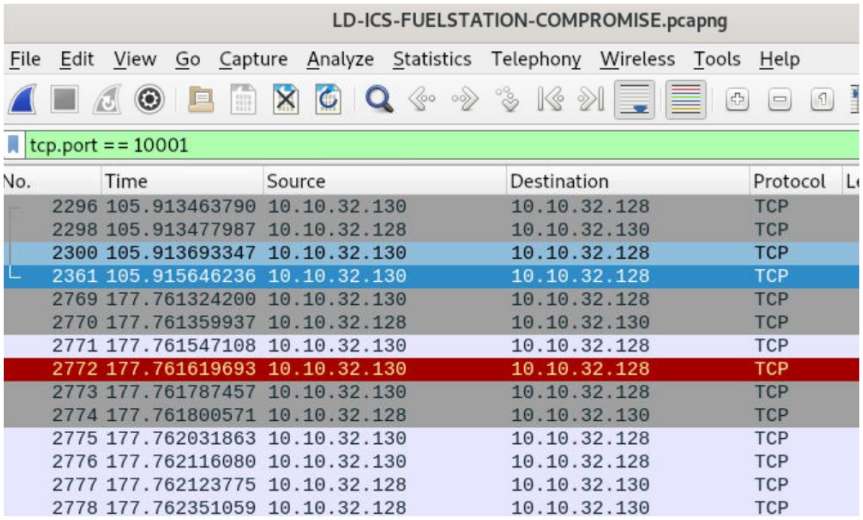
Get unstuck?

✓ **Answer: Veeder**

Question 7 — Facility / Site Name

What is the name of the facility identified in the ICS traffic stream?

> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

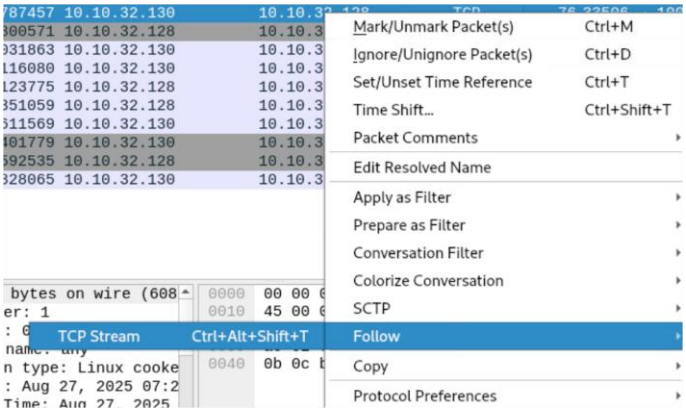


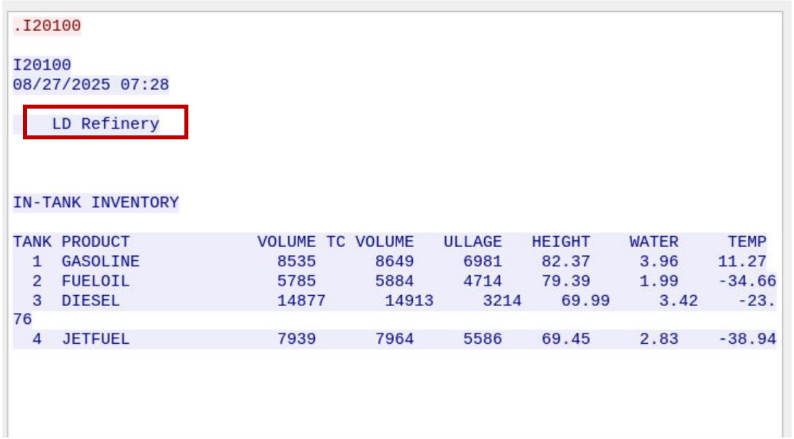
Approach

Veeder-Root ATG protocols embed site metadata in their TCP payloads. Following the TCP stream allows analysts to read ASCII-encoded data including facility names, tank identifiers, and operational parameters — all critical for scope assessment.

Steps Performed

- Applied the filter: tcp.port == 10001
- Right-clicked a packet → Follow → TCP Stream to view decoded payload.
- Located the facility name string within the stream data.





```
.I20100

I20100
08/27/2025 07:28

   LD Refinery


IN-TANK INVENTORY

TANK PRODUCT               VOLUME TC VOLUME   ULLAGE   HEIGHT    WATER     TEMP
  1  GASOLINE                 8535      8649     6981    82.37     3.96    11.27
  2  FUELOIL                  5785      5884     4714    79.39     1.99    -34.66
  3  DIESEL                  14877     14913     3214    69.99     3.42    -23.
76
  4  JETFUEL                  7939      7964     5586    69.45     2.83    -38.94
```

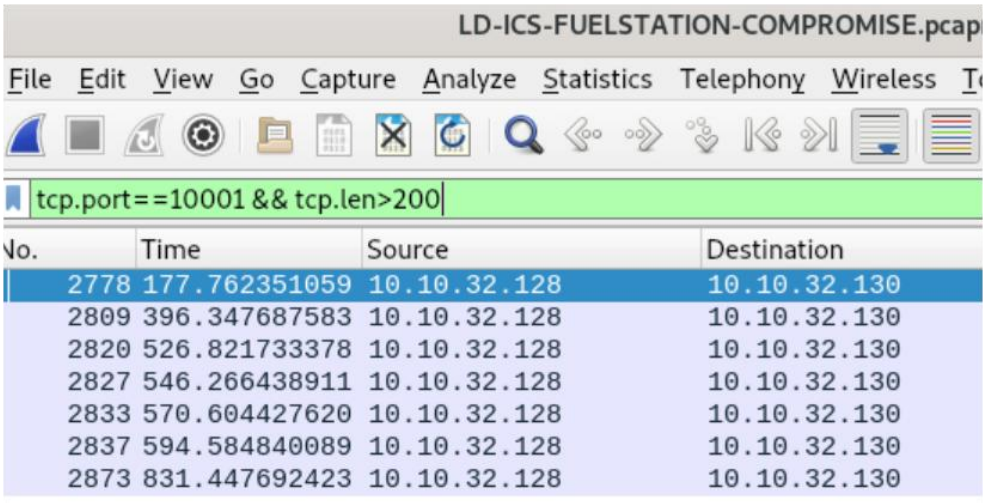


✓ **Answer: LD Refinery**

Question 8 — Tank / Sensor Identifier

What is the identifier of the tank or sensor found in the large payload packet?

> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

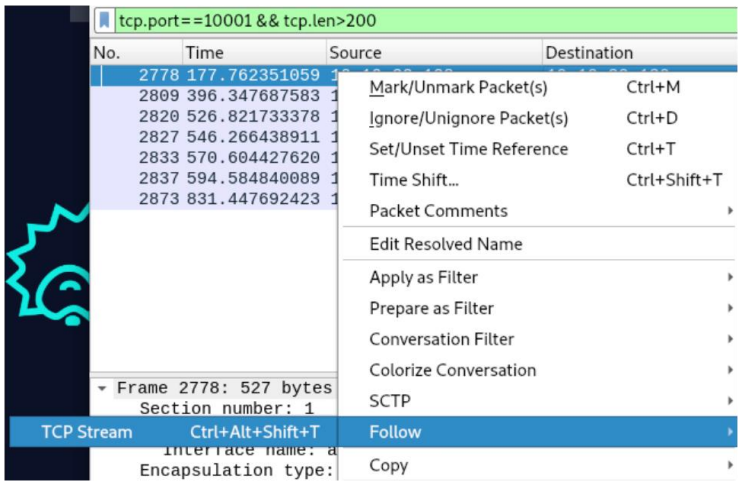


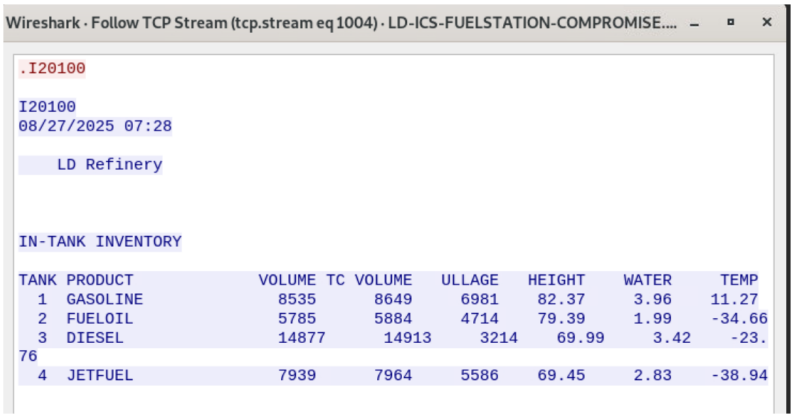
Approach

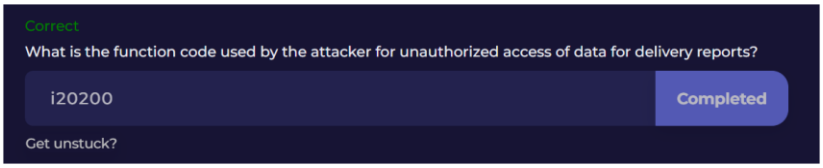
Applying a minimum payload length filter (tcp.len > 200) isolates data-rich packets that contain full ATG reports. These reports include tank IDs, product codes, and level measurements. The stream reveals the exact identifier associated with the monitored fuel tank.

Steps Performed

- Applied the filter: tcp.port == 10001 && tcp.len > 200
- Followed the TCP stream of the first matching packet.
- Extracted the tank/sensor identifier from the decoded stream data.





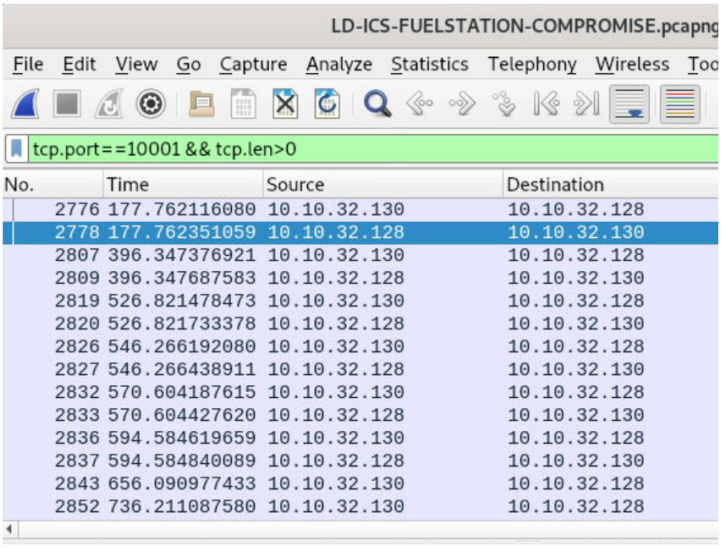


✓ **Answer: I20200**

Question 9 — Scheduled Delivery / Operation Time

What time was a scheduled fuel delivery or operation recorded?

> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

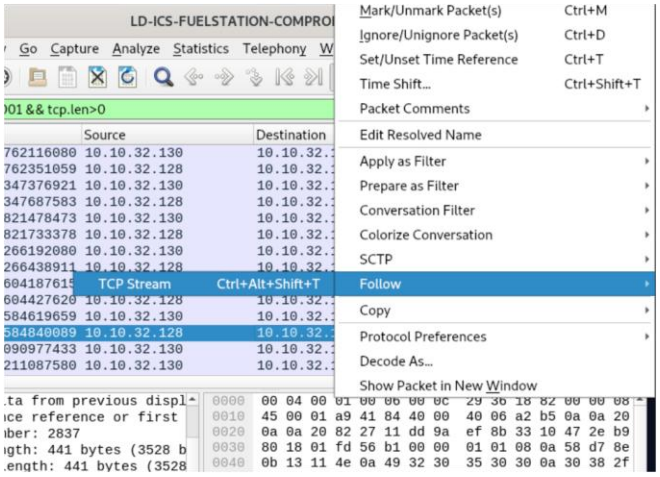


Approach

ICS ATG logs embed timestamped delivery records within their data packets. Analysing the TCP stream of traffic with any payload length (tcp.len > 0) exposes these embedded timestamps, which are essential for correlating physical events with network-level anomalies.

Steps Performed

- Applied the filter: tcp.port == 10001 && tcp.len > 0
- Followed the TCP stream and searched the decoded output for time-formatted strings.
- Located the scheduled operation time recorded in the ATG log.



```
70000000000005
   AMOUNT:                          8535       8649

^AI20300

I20300
08/27/2025 07:34

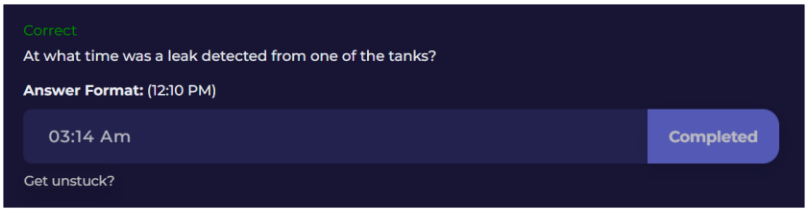
LD Refinery


TANK 1    GASOLINE
    TEST STATUS: OFF
LEAK DATA NOT AVAILABLE ON THIS TANK


TANK 2    FUELOIL
    TEST STATUS: ON
LEAK DETECTED AT 03:14 AM


TANK 3    DIESEL
    TEST STATUS: ON
HEALTHY


TANK 4    JETFUEL
    TEST STATUS: ON
HEALTHY
^AI20400
```



✓ **Answer: 03:14 AM**

Question 10 — Current Fill / Operational Status

What operational status was transmitted in the ICS payload?

> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

Approach

Veeder-Root ATG messages include status codes indicating current tank state — such as filling, idle, or alarm states. Identifying these status strings helps analysts understand what was happening at the physical site during the network capture window.

Steps Performed

- Applied the filter: tcp.port == 10001 && tcp.len > 0
- Followed the TCP stream and examined status flag fields.
- Identified the current operational status string in the decoded data.

```
TANK   PRODUCT                STATUS

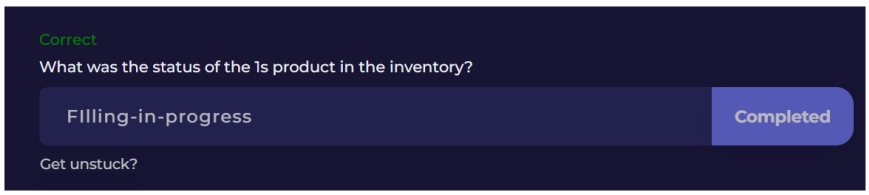
  1    GASOLINE                FILLING-IN-PROGRESS

  2    FUELOIL                 LOW VOLUME ALARM
                               CONTAMINATION WARNING

  3    DIESEL                  NORMAL

  4    JETFUEL                 FULL
^AS60201YOU_ARE_HACKED
^AS60202PAY_US_50_ETH
^AS60203CONTACT_US_AT
^AS60204LEGIONGROUP@PROTONMAIL.COM
^AI20100

I20100
08/27/2025 07:39
```



✓ **Answer: FILLING-IN-PROGRESS**

Question 11 — Delivery Start Timestamp

When did the fuel delivery operation begin?

> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

Approach

ATG protocol streams include both start and end timestamps for delivery events. These timestamps form a forensic timeline of physical-layer events, allowing analysts to establish when fuel manipulation or unauthorised delivery began and correlate it with network-level activity.

Steps Performed

- Applied the filter: tcp.port == 10001 && tcp.len > 0
- Followed the TCP stream and searched for delivery start time markers.
- Extracted the start timestamp from the encoded ATG payload.

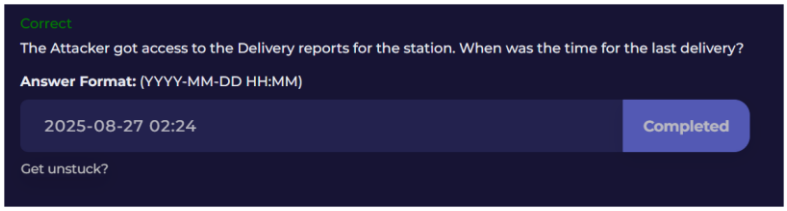
```
DELIVERY REPORT

T 1:GASOLINE
INCREASE    DATE / TIME              GALLONS TC GALLONS WATER   TEMP DEG F   HEI
GHT

      END: 08/27/2025 02:24          8835       8949    3.96       11.27    82.3
7
    START: 08/27/2025 02:14          8235       8349    3.96       11.27    59.3
70000000000005
   AMOUNT:                           8535       8649

^AI20300

I20300
08/27/2025 07:34
```



✓ **Answer: 2025-08-27 02:24**

Question 12 — Delivery End Timestamp

When did the fuel delivery operation end?

> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

Approach

The end timestamp of a delivery window is equally important as the start — together they define the full window of potential tampering or data exfiltration. Comparing start and end times reveals the duration of the suspicious operation.

Steps Performed

- Applied the filter: tcp.port == 10001 && tcp.len > 0
- Continued reviewing the TCP stream for delivery end markers.
- Extracted the end timestamp from the ATG payload.

```
 SHIFT REPORT

SHIFT 4 TIME: 1:00 AM

TANK PRODUCT

  1  GASOLINE                                VOLUME TC VOLUME  ULLAGE  HEIGH
T  WATER   TEMP
SHIFT  1 STARTING VALUES      8535     8649    6981   82.37   3.96    11.27
         ENDING VALUES        9475     9535    7326   135.37  3.96    11.27
         DELIVERY VALUE          1
         TOTALS                7621

^AI20500

I20500
08/27/2025 07:35
```



✓ **Answer: 2025-08-27 07:35**

Question 13 — Fuel Level / Volume Change

What fuel level change (delta) was recorded during the delivery event?

> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

Approach

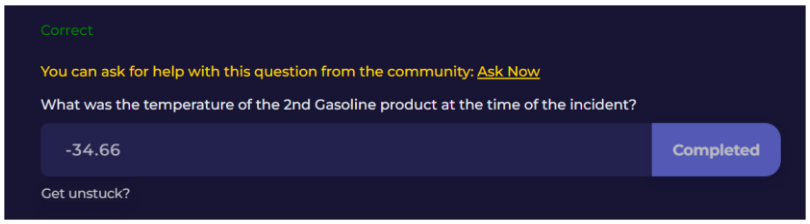
ATG systems continuously record tank level measurements. An anomalous or large level delta can indicate unauthorised fuel extraction or injection. Analysts use this figure to calculate whether recorded volumes match expected delivery manifests.

Steps Performed

- Applied the filter: tcp.port == 10001 && tcp.len > 0
- Followed the TCP stream and identified the volume/level change field.
- Recorded the numeric delta from the ATG data fields.

```
IN-TANK INVENTORY

TANK PRODUCT             VOLUME TC VOLUME   ULLAGE   HEIGHT    WATER     TEM
P
  1  GASOLINE               8535      8649     6981    82.37     3.96    11.2
7
  2  FUELOIL                5785      5884     4714    79.39     1.99    -34.
66
  3  DIESEL                14877     14913     3214    69.99     3.42     -2
3.76
  4  JETFUEL                7939      7964     5586    69.45     2.83    -38.
94
^AI20200
```



✓ **Answer: -34.66**

Question 14 — Fuel Type / Product Classification

What type of fuel or product was stored in the targeted tank?

> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

Approach

The product type field in ATG communications identifies the stored fuel classification. In ICS attack scenarios, this data helps investigators understand the potential safety and environmental impact of an attack, and can be cross-referenced with site manifest records.

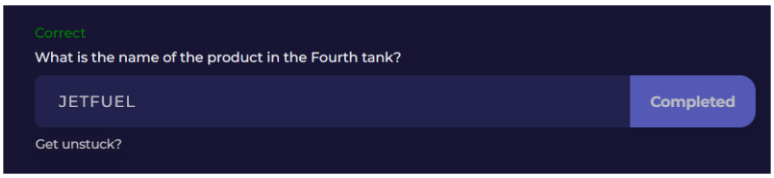
Steps Performed

- Applied the filter: tcp.port == 10001 && tcp.len > 0
- Followed the TCP stream and located the product/fuel type field.
- Identified the fuel classification in the ATG payload.

```
IN-TANK INVENTORY

TANK PRODUCT             VOLUME TC VOLUME   ULLAGE   HEIGHT    WATER     TEM
P
  1  GASOLINE               8535      8649     6981    82.37     3.96    11.2
7
  2  FUELOIL                5785      5884     4714    79.39     1.99    -34.
66
  3  DIESEL                14877     14913     3214    69.99     3.42     -2
3.76
  4  JETFUEL                7939      7964     5586    69.45     2.83    -38.
94
^AI20200

I20200
08/27/2025 07:34
```



✓ **Answer: Jet Fuel**

Question 15 — DHCP Hostname / Site Identifier

What is the hostname of the compromised device as identified via DHCP?

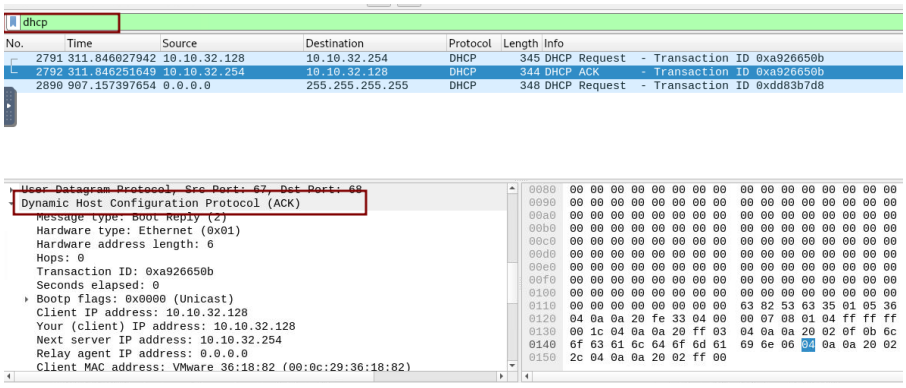
> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

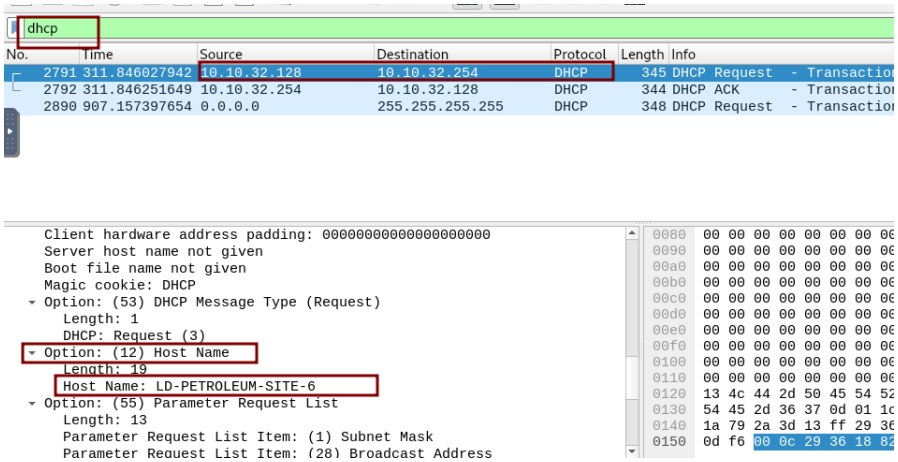
Approach

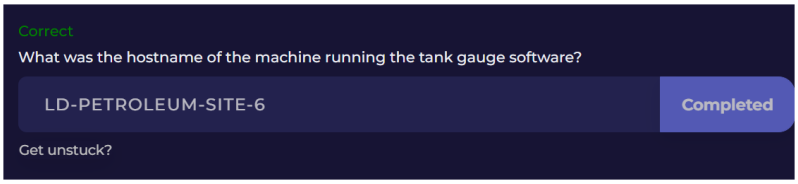
DHCP Option 12 carries the hostname that a device broadcasts when requesting a network address. This hostname is a critical pivot point in endpoint identification — matching it to known asset inventories helps confirm which physical device was compromised.

Steps Performed

- Applied the Wireshark display filter: dhcp
- Expanded the Dynamic Host Configuration Protocol section in the packet details.
- Located Option 12 (Host Name) and read the value.







✓ **Answer: LD-PETROLEUM-SITE-6**

Question 16 — Ransom Demand / Extortion Amount

What ransom demand was embedded in the attacker's communication?

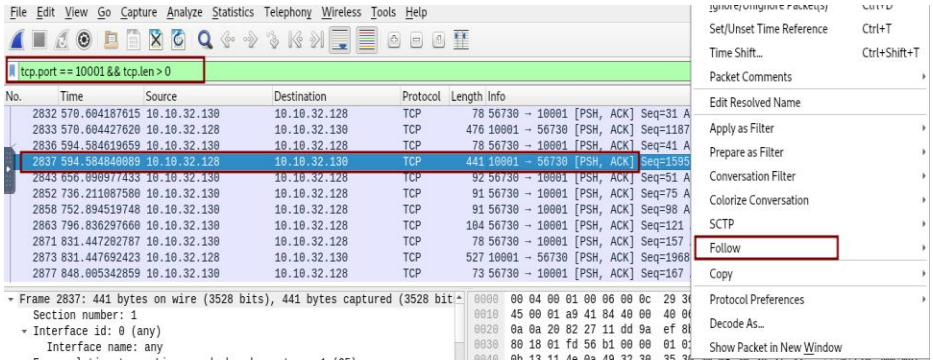
> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

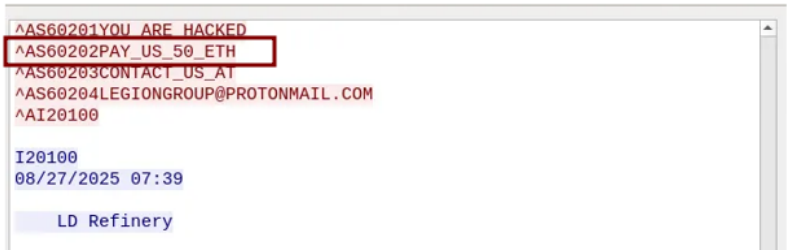
Approach

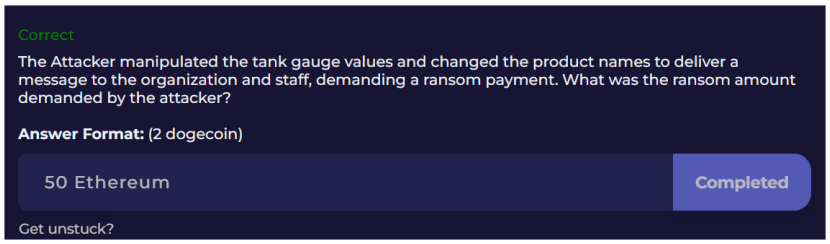
Sophisticated ICS threat actors often embed ransom demands or extortion messages within the same communication channels they use for C2. Reviewing the full TCP stream content for human-readable strings can reveal attacker intent, communication style, and negotiation demands.

Steps Performed

- Applied the filter: tcp.port == 10001 && tcp.len > 0
- Followed the TCP stream and scanned for natural-language attacker messages.
- Located the extortion demand embedded within the stream data.







✓ **Answer: 50 Ethereum**

Question 17 — Attacker Contact / C2 Email Address

What email address did the attacker embed for contact or C2 communication?

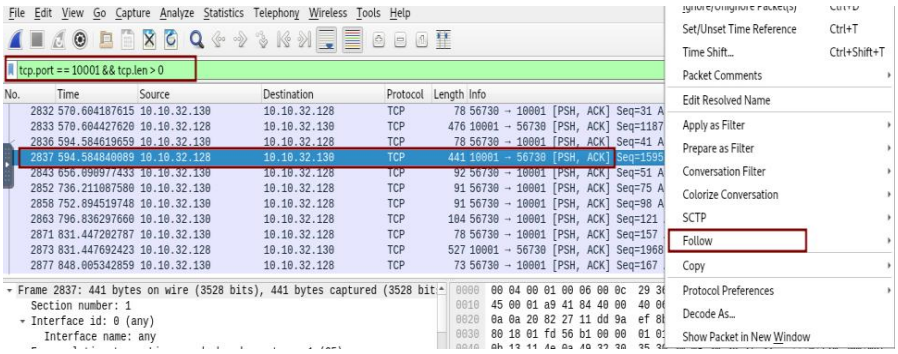
> *Filter Used: See Steps Performed below for the exact Wireshark display filter applied.*

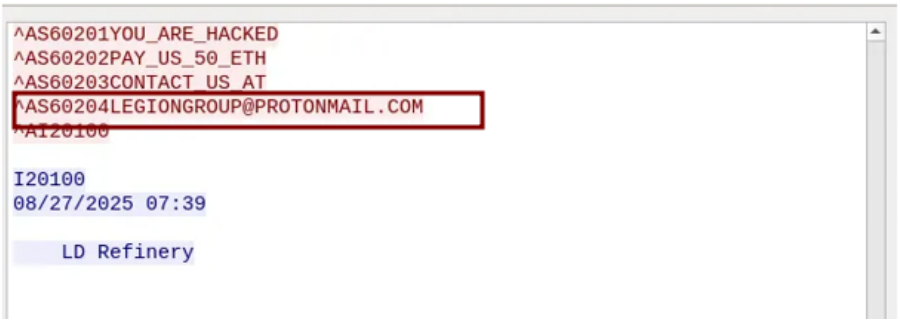
Approach

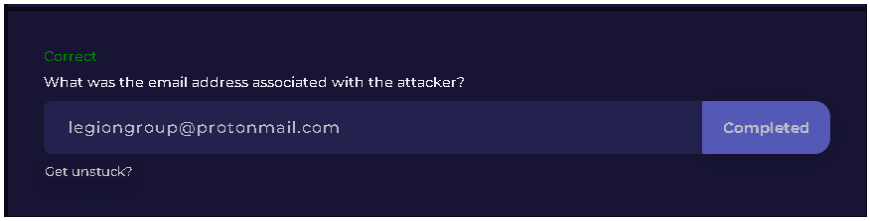
Threat actors frequently embed email addresses — especially anonymous ProtonMail addresses — in their ransom notes or C2 channels for victim communication. Extracting this indicator of compromise (IoC) enables threat intelligence teams to track the threat actor across other incidents.

Steps Performed

- Applied the filter: tcp.port == 10001 && tcp.len > 0
- Followed the TCP stream and searched for email-format strings.
- Extracted the attacker's ProtonMail contact address from the stream.







✓**Answer: legiongroup@protonmail.com**

